

Quick Check

Lesson 10: Presenting Our Final Explanation and Sharing Seedlings with Kitale

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 10

Mr. Wafula planted maize and beans on the same day, in the same field, after the same 25 mm of rain. Fewer bean seeds than maize seeds germinated, and they grew differently. Which explanation is best supported by the evidence?

- A** Healthy-looking seeds always germinate, so the differences are just chance.
- B** The beans were planted on a different day from the maize.
- C** The differences come from seed structure, seed condition (such as dormancy and viability) and the environment (such as waterlogging and oxygen).
- D** The rain fell more on the beans than on the maize.

Out of 100 Rosecoco bean seeds planted by Mr. Wafula, 80 germinated. The seeds that failed were: 8 with hard seed coats (dormant), 6 in the waterlogged patch, 4 non-viable and 2 damaged by pests. How many seeds failed to germinate in total, and does this agree with the 80 that germinated?

- A 20 failed, which agrees with 80 germinating
- B 22 failed, so 78 should have germinated
- C 12 failed, which agrees with 88 germinating
- D 18 failed, so 82 should have germinated

A learner puts 20 bean seeds in each of two jars kept at 25 °C. Jar 1 has moist cotton wool and air, and 18 seeds germinate. Jar 2 has the same temperature and water but the seeds are fully covered in water with no air, and only 1 seed germinates. What does this show?

- A** Seeds need soil to germinate.
- B** Seeds need oxygen for respiration to germinate, which is why waterlogged seeds fail.
- C** Seeds germinate best when they have too much water.
- D** Seeds need light to germinate.

Eight bean seeds looked healthy but did not germinate in moist, warm soil. After the hard seed coat was scratched (scarification), 7 of the 8 germinated. What is the best explanation?

- A** The seeds were non-viable (dead) from the start.
- B** The seeds had too much oxygen before they were scratched.
- C** The seeds were damaged by pests, which scarification repaired.
- D** The seeds were dormant because the hard testa stopped water from entering.

Which statement correctly compares how maize and bean seedlings emerge from the soil?

- A** Maize is hypogeal: its scutellum stays underground while the coleoptile pierces the soil. Beans are epigeal: their two cotyledons are raised above ground.
- B** Both maize and beans are epigeal because both have a plumule.
- C** Both maize and beans are hypogeal because both are planted 3 cm deep.
- D** Maize is epigeal: its cotyledons are lifted above the ground. Beans are hypogeal.

A bean plant with its shoot tip left on grew 0 branches. After the tip was removed at Day 20, it grew 3 branches. What explains this?

- A** Removing the tip stopped the plant from respiring, so the side buds grew.
- B** Auxin made in the shoot tip inhibits the side (lateral) buds, so removing the tip lets them grow into branches.
- C** Gibberellins from the roots only work after the tip is cut off, so the side buds die.
- D** Absciscic acid from the tip makes the plant grow taller, so branches cannot form.

Maize grew to 125 cm and beans to 44 cm by Day 42. In which part of the plant does the primary growth that lengthens shoots and roots take place?

A In the apical meristems at the tips of the shoots and roots

B In the testa and micropyle of the seed

C In the mature leaf cells that are fully expanded

D In the endosperm stored in the seed

The school nursery sells 60 aloe vera-treated seedlings at KES 20 each. The polythene tubes and soil cost KES 300. What is the profit?

- A KES 900
- B KES 1,200
- C KES 300
- D KES 1,500

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).